

Adaptive Multi-Agent Cyber Defense Framework Using GraphRAG and Large Language Models

Member 1 — Threat Detection & Real-Time Event Processing

Final Results Report

Generated: September 19, 2026

Model Comparison — Validation Set

dataset	model	accuracy	precision	recall	f1_weighted	f1_macro	latency_ms
CIC-IDS2018	Logistic Regression	0.9995	0.9995	0.9995	0.9995	0.9990	0.00023
CIC-IDS2018	Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000	0.00789
CIC-IDS2018	XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000	0.00752
TON_IoT	Logistic Regression	0.9872	0.9880	0.9872	0.9871	0.9696	0.00025
TON_IoT	Random Forest	0.9999	0.9999	0.9999	0.9999	0.9998	0.02234
TON_IoT	XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000	0.02362

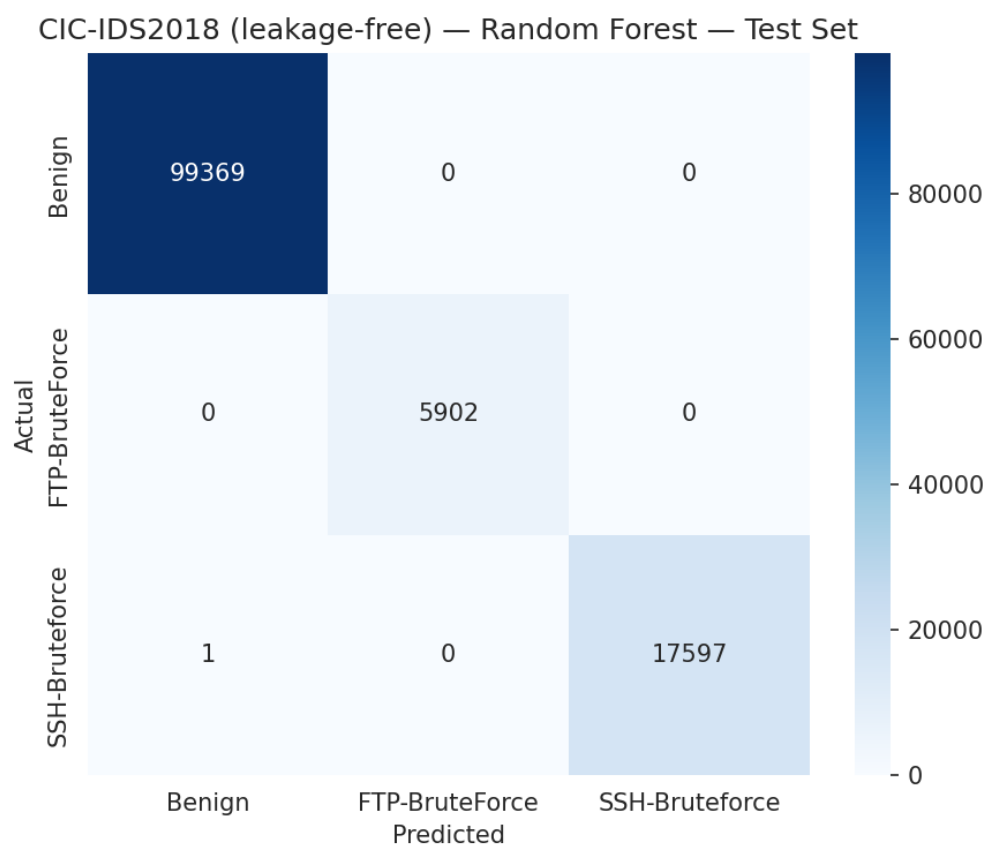


Figure 1: CIC-IDS2018 Confusion Matrix (Test Set, leakage-free)

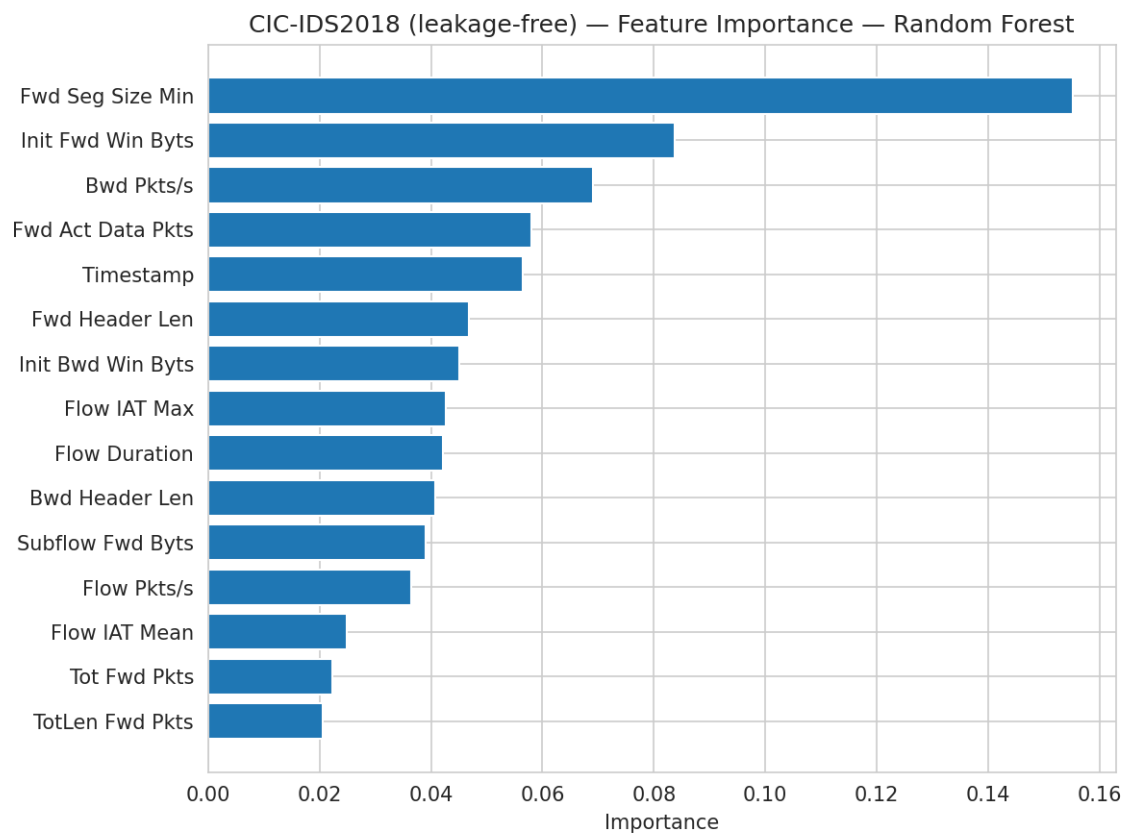


Figure 2: CIC-IDS2018 Feature Importance

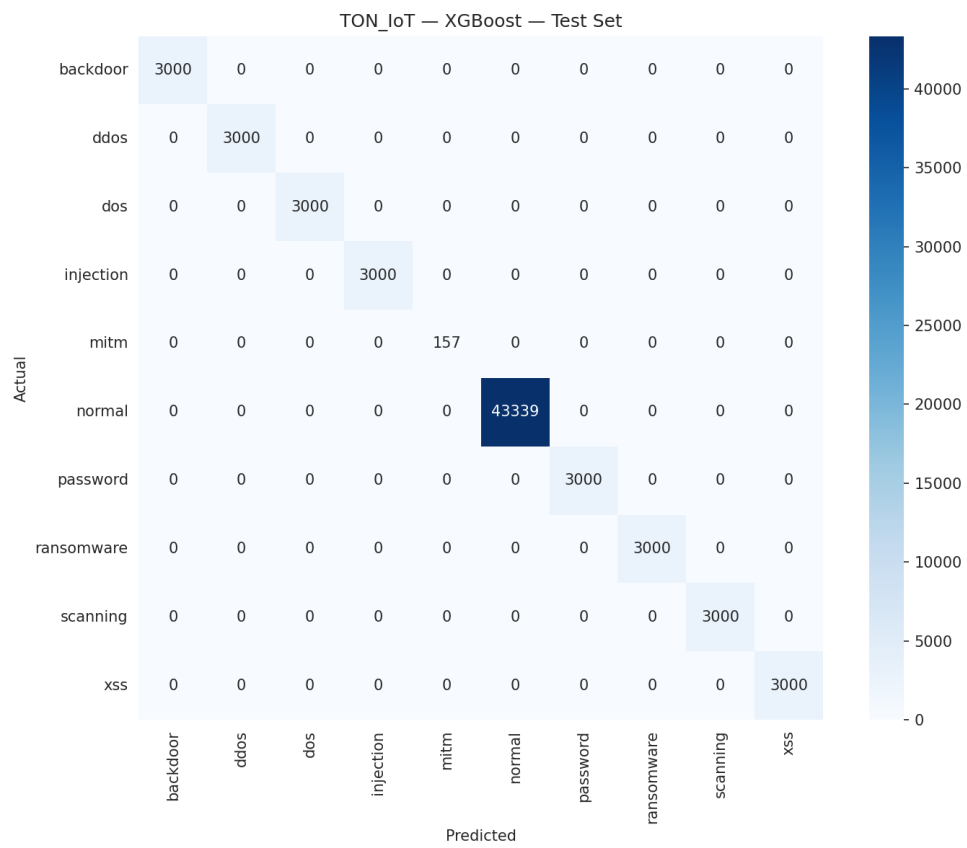


Figure 3: TON_IoT Confusion Matrix (Test Set)

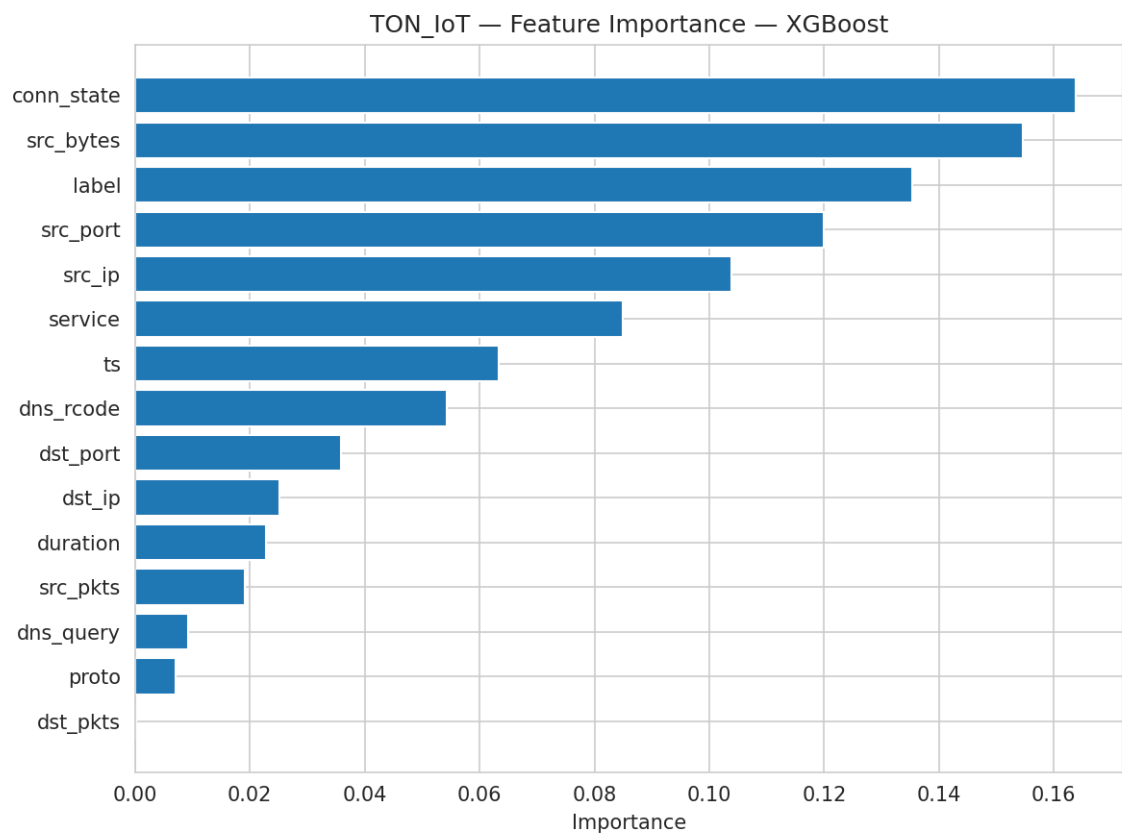


Figure 4: TON_IoT Feature Importance

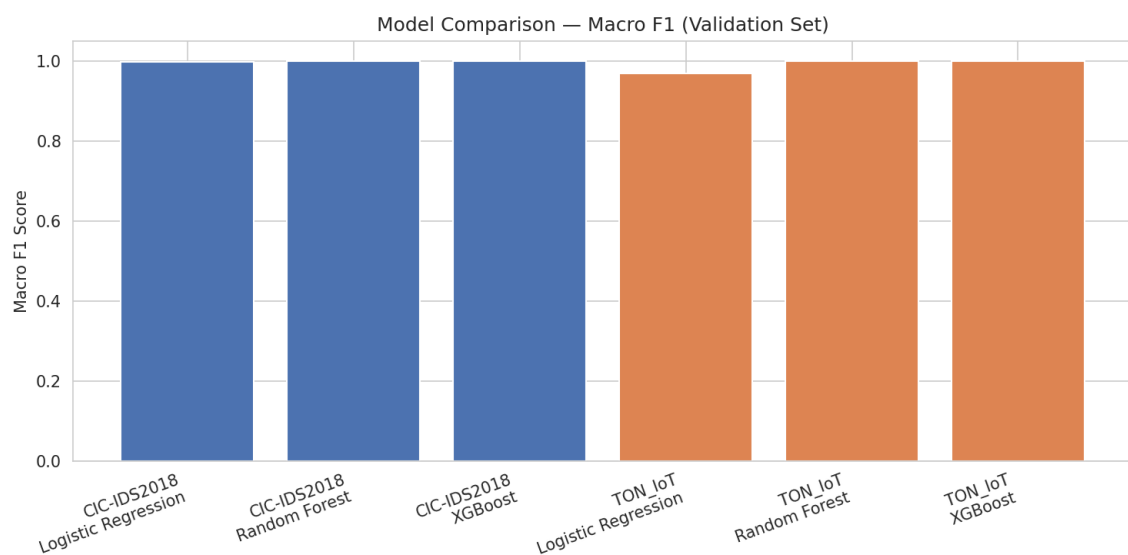


Figure 5: Model Comparison — Macro F1 Across All Models

Full Results Summary

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MEMBER 1 – FINAL RESULTS SUMMARY

Threat Detection and Real-Time Event Processing

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--- DATASET 1: CSE-CIC-IDS2018 ---

File: 02-14-2018.csv

Raw shape: 1048575 rows x 80 columns

--- DATASET 2: TON_IoT ---

File: Train_Test_Network.csv

Raw shape: 461043 rows x 45 columns

--- PREPROCESSING SUMMARY ---

CIC-IDS2018: dropped 225628 duplicate rows, 3821 rows with missing/infinite values, 14 redundant correlated features

Final shape: (819126, 65)

TON_IoT: dropped 11071 duplicate rows, 2 redundant correlated features

Final shape: (449972, 43)

CIC-IDS2018 class distribution:

Label	
Benign	662458
SSH-Bruteforce	117322
FTP-BruteForce	39346

TON_IoT class distribution:

type	
normal	288929
scanning	20000
dos	20000
injection	20000
ddos	20000
password	20000
xss	20000
ransomware	20000
backdoor	20000
mitm	1043

--- SPLITS ---

CIC-IDS2018 train/val/test: 573388 / 122869 / 122869

TON_IoT train/val/test: 2022500 (post-SMOTE) / 67496 / 67496

--- MODEL COMPARISON (validation set) ---

dataset	model	train_time_s	latency_ms	accuracy	precision	recall	f1_weighted	f1_macro
CIC-IDS2018	Logistic Regression	6.41	0.00023	0.999528	0.999529	0.999528	0.999528	0.999035
CIC-IDS2018	Random Forest	210.32	0.00789	1.000000	1.000000	1.000000	1.000000	1.000000
CIC-IDS2018	XGBoost	71.44	0.00752	1.000000	1.000000	1.000000	1.000000	1.000000
TON_IoT	Logistic Regression	1272.78	0.00025	0.987184	0.987978	0.987184	0.987130	0.969554
TON_IoT	Random Forest	595.11	0.02234	0.999911	0.999911	0.999911	0.999911	0.999800
TON_IoT	XGBoost	485.77	0.02362	1.000000	1.000000	1.000000	1.000000	1.000000

Best model – CIC-IDS2018: Random Forest (macro F1 = 1.0000)

Best model – TON_IoT: XGBoost (macro F1 = 1.0000)

--- FINAL TEST SET RESULTS ---

CIC-IDS2018 (Random Forest, leakage-free – 'Dst Port' removed):

	precision	recall	f1-score	support
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Benign	1.00	1.00	1.00	99369
FTP-BruteForce	1.00	1.00	1.00	5902
SSH-Bruteforce	1.00	1.00	1.00	17598

accuracy			1.00	122869
macro avg	1.00	1.00	1.00	122869
weighted avg	1.00	1.00	1.00	122869

TON_IoT (XGBoost):

	precision	recall	f1-score	support
backdoor	1.00	1.00	1.00	3000
ddos	1.00	1.00	1.00	3000
dos	1.00	1.00	1.00	3000

injection	1.00	1.00	1.00	3000
mitm	1.00	1.00	1.00	157
normal	1.00	1.00	1.00	43339
password	1.00	1.00	1.00	3000
ransomware	1.00	1.00	1.00	3000
scanning	1.00	1.00	1.00	3000
xss	1.00	1.00	1.00	3000
accuracy			1.00	67496
macro avg	1.00	1.00	1.00	67496
weighted avg	1.00	1.00	1.00	67496

Top CIC-IDS2018 features (post leakage-fix): [('Fwd Seg Size Min', 0.1551), ('Init Fwd Win Byts', 0.0837), ('Bwd

Top TON_IoT features: [('conn_state', 0.1638), ('src_bytes', 0.1547), ('label', 0.1354), ('src_port', 0.1199), ('

--- INFERENCE LATENCY (per-sample, validation set) ---

CIC-IDS2018 – Logistic Regression: 0.00023 ms/sample

CIC-IDS2018 – Random Forest: 0.00789 ms/sample

CIC-IDS2018 – XGBoost: 0.00752 ms/sample

TON_IoT – Logistic Regression: 0.00025 ms/sample

TON_IoT – Random Forest: 0.02234 ms/sample

TON_IoT – XGBoost: 0.02362 ms/sample

--- LEAKAGE DIAGNOSTIC (post-fix, for methodology section) ---

Top single-feature predictive accuracy on CIC-IDS2018 (leakage-free feature set): [('Fwd Seg Size Min', 0.996506)

Note: 'Dst Port' was identified as a leakage source (99.95% standalone accuracy) and removed prior to final training. See methodology section.

KNOWN DATASET LIMITATIONS

The near-perfect classification performance observed here (macro F1 = 1.0000 for Random Forest on CSE-CIC-IDS2018, and 1.0000 for XGBoost on TON_IoT) should be interpreted with caution rather than as evidence of production-ready detection capability.

CSE-CIC-IDS2018: The single-feature diagnostic above shows individual flow features achieving near-perfect standalone predictive accuracy (top: 'Fwd Seg Size Min' at 99.65%). An earlier pipeline version also found 'Dst Port' alone achieved 99.95% accuracy (an artifact of brute-force traffic using fixed ports 21/22), which was removed before final training. This pattern is consistent with published findings of systematic errors in the CIC-IDS dataset family:

- Engelen, Rimmer & Joosen (2021), "Troubleshooting an Intrusion Detection Dataset: the CICIDS2017 Case Study," IEEE S&P Workshops, found >25% of flows are "meaningless artefacts" and confirmed the issue persists in CSE-CIC-IDS2018.
- Liu, Engelen, Lynar, Essam & Joosen (2022), "Error Prevalence in NIDS Datasets: A Case Study on CIC-IDS-2017 and CSE-CIC-IDS-2018," IEEE CNS, further characterized these errors.
- Cantone et al. (2024) found models trained/evaluated on CSE-CIC-IDS2018 achieve near-perfect in-dataset accuracy but near random-chance performance on external datasets, indicating in-dataset results substantially overstate real-world generalization.

TON_IoT: A synthetic/lab-generated dataset; the single-feature check found no feature exceeding 51.5% standalone accuracy, so near-100% scores here likely reflect low real-world noise in controlled-generation data rather than a specific leakage source. Real-world performance is expected to be lower.

Implication: these results validate the detection pipeline's architecture (cleaning, leakage diagnosis, model comparison, real-time scoring) rather than claiming generalizable real-world accuracy. Future work should use cross-dataset validation (train on one dataset, test on an independently generated one) for a realistic generalization estimate.